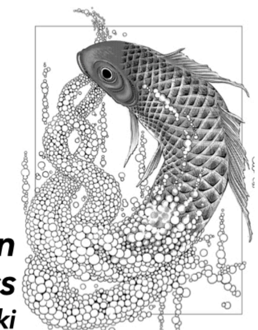
The origin of ferromagnetism and the Hubbard model

part 1: introduction



Advanced Topics in Statistical Physics
by Hal Tasaki

Illustration © Mari Okazaki 2020. All Rights Reserved

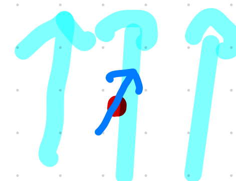
why are there ferromagnets?

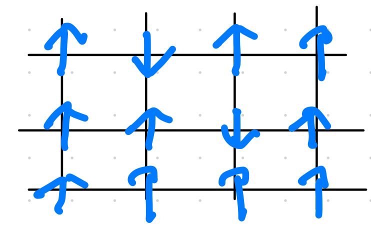
=

solids with macroscopic magnetic moments

"modern" microscopic picture

Weiss 1907 molecular field





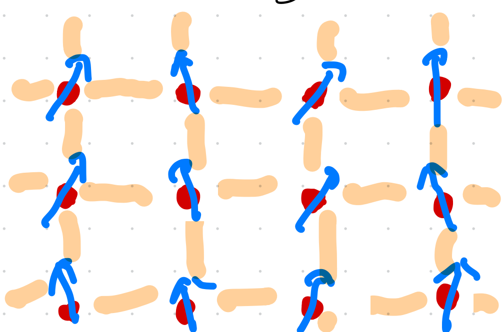
Lenz 1920, Ising 1927 simple microscopic model / spin system!

Heisenberg 1928

magnetic moment of electrons localized around atoms

+

exchange interaction originating from quantum mechanics



may apply only to ferromagnets that are insulators

NOT to iron or neodymium magnet

insulating ferromagnet

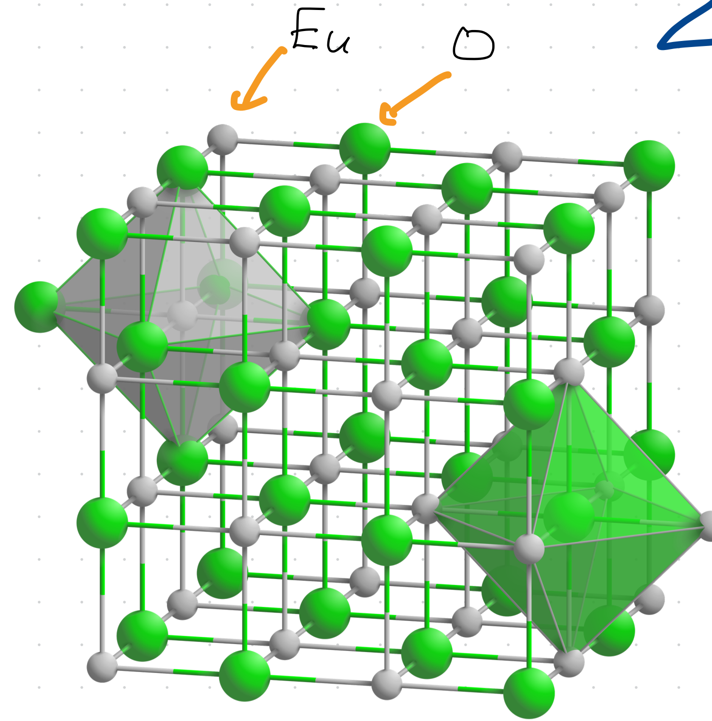
Z

EuO

magnetic moment comes mainly from electrons localized at Eu 4f state.

$$S = \frac{7}{2}$$

effective exchange interaction mediated by Eu 5d and 6s states



Crystal structure of NaCl with coordination polyhedra by Goran tek-en (from Wikipedia)

the picture is essentially different for metallic ferromagnetism

Heisenberg 1928

- symmetry of fermions + Coulomb repulsion → exchange interaction
- perturbative argument for two electrons localized at neighboring atoms

how ferromagnetism emerge in a macroscopic solid?

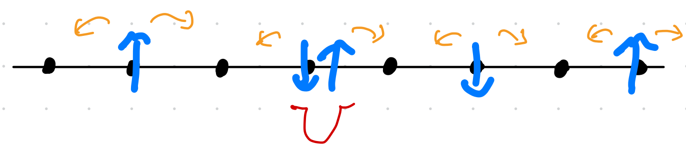
Hubbard model

Kanamori 1963, Gutzwiller 1963, Hubbard 1963

tight-binding description of electrons in relevant states

\+

on-site Coulomb interaction U



- originally introduced to study (mainly) ferromagnetism
- describes various phenomena in solids; antiferromagnetism, ferrimagnetism, Mott insulator, superconductivity, ...

Contents